

Krish Shrivastav

LinkedIn | GitHub | Portfolio | Pune, Maharashtra | 8983269528 | srivastavkrish15@gmail.com

PROFESSIONAL SUMMARY

Third-year BCA student with strong programming and software engineering fundamentals. Built systems and full-stack projects including a database engine from scratch and a real-time collaborative application using WebSockets and CRDTs. Skilled in C++, Python, JavaScript, React, Node.js, PostgreSQL, REST APIs, and Git.

EDUCATION

Bachelor of Computer Applications (BCA) | 2024 – 2027

Sinhgad College of Science, Savitribai Phule Pune University (SPPU)

SGPA: FY – Sem 1: 9.36, Sem 2: 8.64 | SY – Sem 3: 8.36, Sem 4: 9.27 | TY – Current Year

TECHNICAL SKILLS

Languages	C, C++, Python, Java, JavaScript
Web	HTML, CSS, React.js, Node.js, Flask
Databases	SQL, PostgreSQL, Oracle
Tools & Concepts	Git, GitHub, WebSockets, REST APIs

PROJECTS

Collab Whiteboard – Real-Time Collaborative Canvas

Live Demo | GitHub

- Built a real-time collaborative whiteboard using React, Konva.js, Node.js, and WebSockets, enabling synchronized canvas editing across multiple connected users.
- Implemented multi-user sync using Yjs CRDTs, with live cursor tracking on a separate ephemeral channel.
- Added per-user undo history and persistent shape storage to disk.

MiniDB – Key-Value Database Engine

Live Demo

- Built a key-value database engine from scratch in Python with no third-party dependencies in the core.
- Designed a durable append-only storage log with CRC32 integrity checks, crash recovery, and an O(1) in-memory primary index.
- Implemented a self-balancing B+ tree for secondary indexes supporting O(log n) equality and range queries.
- Implemented TTL-based key expiry, atomic transactions, thread-level locking, process-level file locking, and automatic log compaction; exposed the engine through TCP and Flask HTTP interfaces.

Personal Portfolio Website

krishshrivastav.in

- Developed a responsive personal website to present academic projects, technical skills, and profile.
- Implemented clean layouts and interactive elements using HTML, CSS, and JavaScript, focused on usability and performance.

ACTIVITIES & INTERESTS

Recreational sports – cricket, football, badminton. Casual gaming – Valorant, Mobile Legends: Bang Bang (MLBB), BGMI, CS:GO.

LANGUAGES

Hindi, English, Marathi